[Lecture 1] Exploring Data

CS5701 - Quantitative Data Analysis

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Week 1 - Learning Outcomes

This week after the lecture, the lab and the independent practice you should be:

- Explain why statistics is needed and important
- Able to determine the different data types
- Suggest and use appropriate methods for numerical and graphical exploration depending on the data type
- Able to use R Studio to read in data and run some numerical and graphical exploration

What is Statistics?

> *"Statistics is concerned with* **collecting, analysing** *and* **interpreting** *data in the best possible way, where the meaning of 'best' depends on the particular circumstances of the practical situation"*

Quote from C. Chatfield, "Problem Solving A Statistician's Guide"

Why is Statistics Important?

- There is great potential for using data to better understand the world, make better judgements and Statistics gives us a way of achieving this
- For example Statistics can help us determine if a change in a value or measure is real or just an artefact of randomness - does a treatment work?
- Statistical methods are important for data science

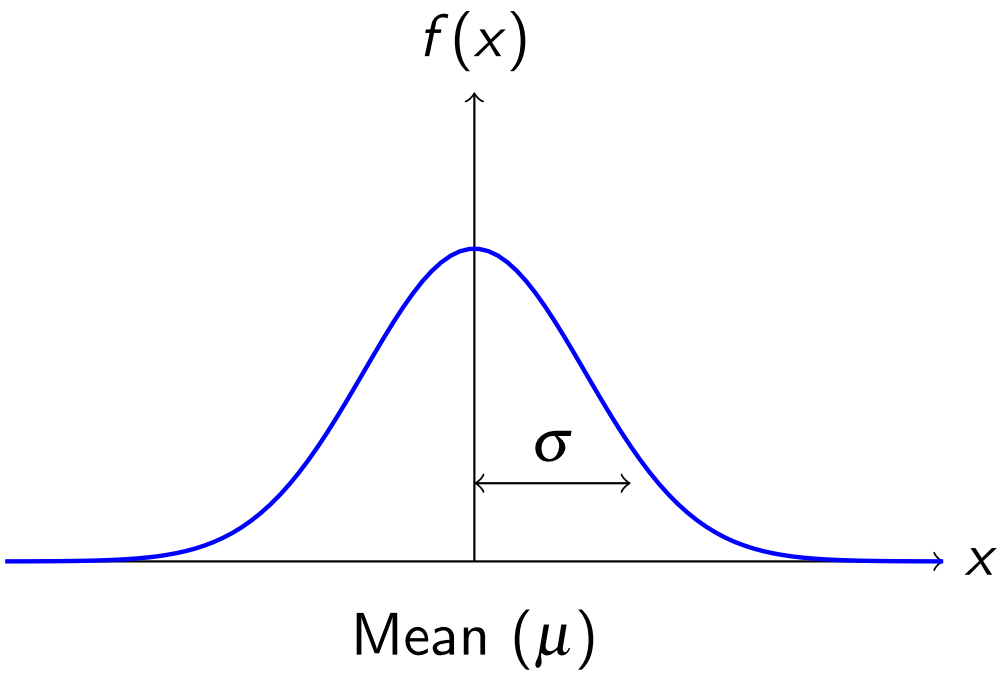
But what about AI and GenAI

Do we still need statistics?

- Behind every AI decision is **probability** and **statistics**
- For example "How does AI recognise Cats?"

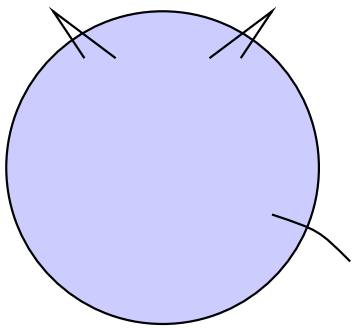


Building Blocks: Mean and Standard Deviation

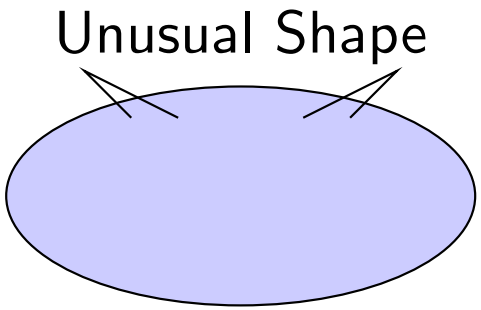


- Mean (μ): The "average" or center of our data
- Standard Deviation (σ): How spread out the data is

How AI Learns "Normal" Patterns



Normal Cat

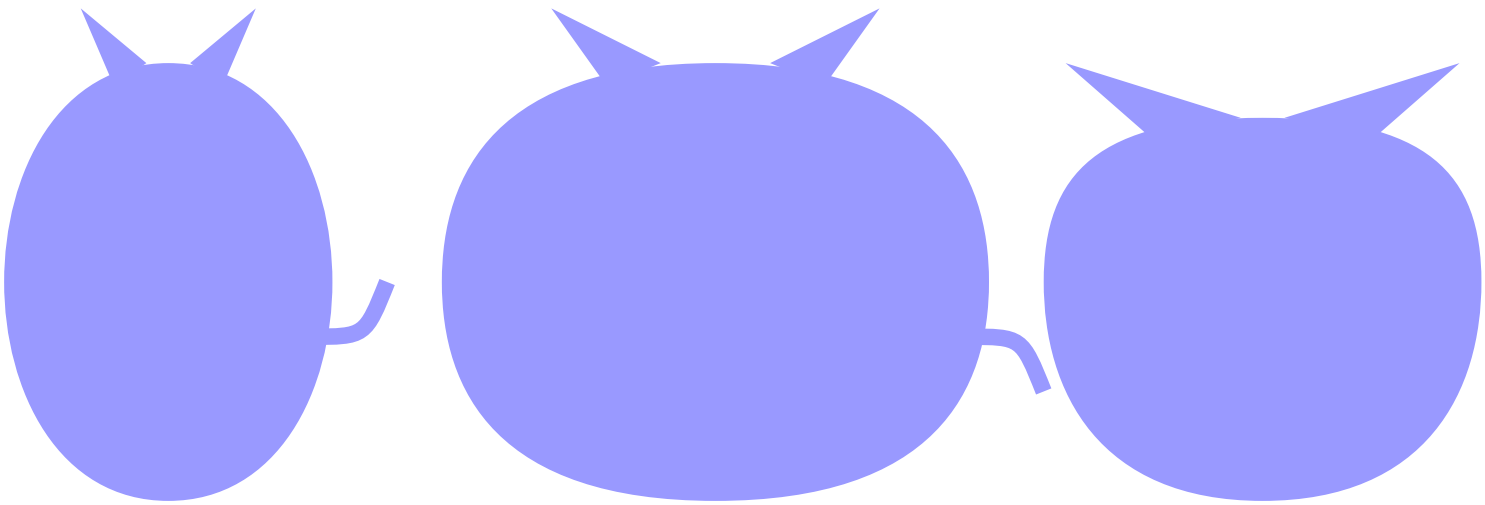


- AI learns from thousands of examples
- Calculates "typical" measurements
- Uses **statistics** to identify unusual cases

How Does AI Recognise Cats?

- AI learns from thousands of cat images
- It looks for common patterns and measurements
- Uses statistics to determine what's "normal"

Three Types of Shapes

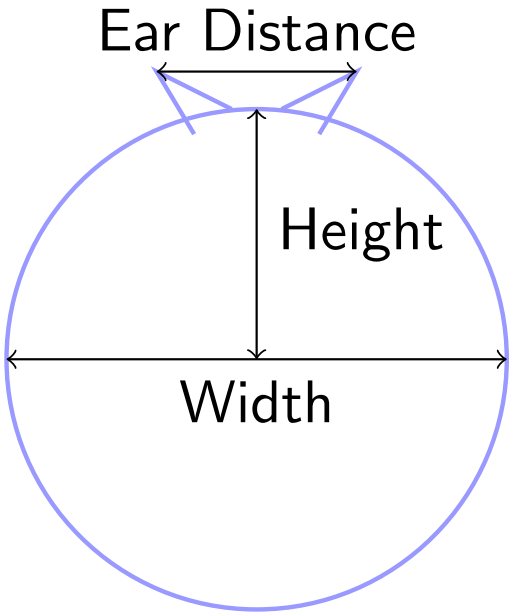


Normal Cat

Unusual Shape

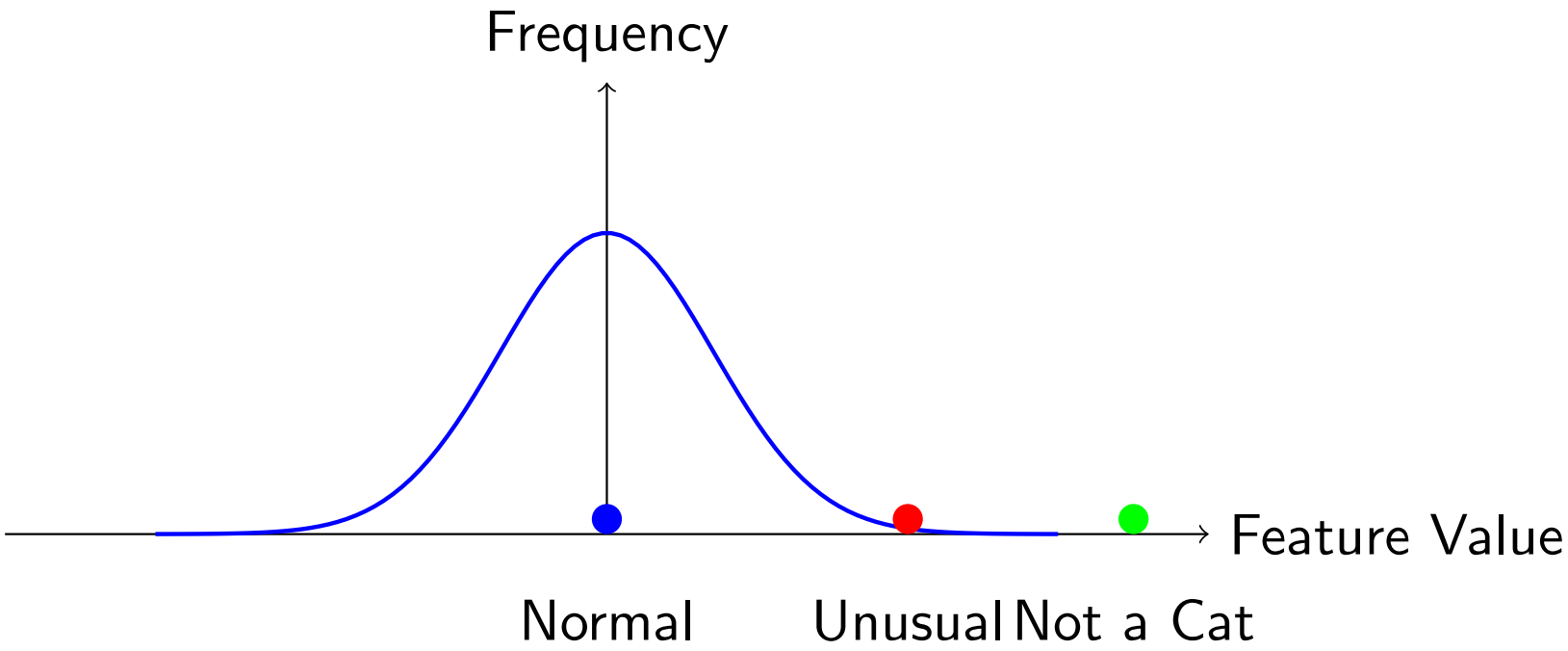
Not a Cat

Statistical Features AI Looks For



- Key measurements (feature):
  - Distance between ears
  - Body proportions (height/width ratio)
  - Ear shape and angle

Statistical Distribution of Features



- Normal cats cluster around the mean
- Unusual shapes appear at the edges
- Non-cats fall outside typical range

How AI Makes Decisions

- Calculates probability scores
- Compares with learned patterns
- Makes confidence-based decisions

Shape	**Confidence**
Normal Cat	95%
Unusual Shape	60%
Not a Cat	5%

Why This Matters

- Shows how AI uses statistics to:
  - Learn patterns
  - Handle variations
  - Make decisions
  - Generate the next word
- Same principles apply to:
  - Face recognition
  - Medical imaging
  - Quality control
  - LLMs

One More Reason Statistics Matters

- LLMs like ChatGPT or Claude generate text by predicting a **probability distribution** over the next word - the same probability and statistics ideas we just used to spot an unusual cat
- When researchers build these models, they don't just assume a design choice helped - they test it, by removing or changing one part of the model and measuring what happens to performance
- This is called an **ablation study**
- Sound familiar? It's the same question we asked earlier: is a change real, or just noise?

We will come back to ablation studies directly towards the end of the semester.

Types of Analysis

There are two main types of analysis:

- **Descriptive**: describing the data by using numerical summaries or graphical outputs
- **Inferential**: goes beyond describing the data by using the information from a sample of data to make a conclusion about a larger population.

Inferential Statistics - Population vs. sample

In **Inferential statistics** the aim is to use the information from a sample to *infer* about the larger population. For example to determine the mean height of a 12 year old child, it is not necessary to measure all 12 year old children in the UK.

- **Population** of interest: this is the entire group of items or individuals about which we want to estimate its parameters. In the example this is all the 12 year old children in the UK this year.
- **Sample** a smaller set of data from the population of interest, that is available to be analysed and upon which the parameters of the populations can be inferred. In the example these can be a set of 10 classes of 12 year olds in London.

This lecture wont focus on inference, we will cover this in later lectures.

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Descriptive Statistics

- Types of variables
- How to summarise each type numerically
- How to visualise each type using plots

Descriptive Statistics - Types of Variables

- **Categorical** - records which one of a list of possible categories or attributes is observed for a particular sampling unit. When there are two possible categories it is known as **binary**
- **Numerical** - takes a numerical value, which can be on a discrete, ordinal or continuous scale. Sometimes called **interval** or **continuous**

Note: that a variable can also be referred to as an attribute, a column (for example in a spreadsheet) or a vector of data.

Worms data

This table has the first 10 rows of the Worms Data set from Crawley's book (Ch 2):

	Field.Name	Area	Slope	Vegetation	Soil.pH	Damp	Worm.density
1	Nashs.Field	3.60	11	Grassland	4.10	FALSE	4
2	Silwood.Bottom	5.10	2	Arable	5.20	FALSE	7
3	Nursery.Field	2.80	3	Grassland	4.30	FALSE	2
4	Rush.Meadow	2.40	5	Meadow	4.90	TRUE	5
5	Gunness.Thicket	3.80	0	Scrub	4.20	FALSE	6
6	Oak.Mead	3.10	2	Grassland	3.90	FALSE	2
7	Church.Field	3.50	3	Grassland	4.20	FALSE	3
8	Ashurst	2.10	0	Arable	4.80	FALSE	4
9	The.Orchard	1.90	0	Orchard	5.70	FALSE	9
10	Rookery.Slope	1.50	4	Grassland	5.00	TRUE	7

What types of variables are these?

Worms data

	Field.Name	Area	Slope	Vegetation	Soil.pH	Damp	Worm.density
1	Nashs.Field	3.60	11	Grassland	4.10	FALSE	4
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- `Field.Name, Vegetation, Damp` are **categorical**.
- `Area, Slope, Worms` are **numerical**

When you use data make sure to check the metadata or data description - sometimes a number actually can represent a category....

Another example

What type of variables are `extra` and `group`?

	extra	group	ID
1	0.70	1	1
2	-1.60	1	2
3	-0.20	1	3
4	-1.20	1	4
5	-0.10	1	5
6	3.40	1	6
7	3.70	1	7
8	0.80	1	8
9	0.00	1	9
10	2.00	1	10
11	1.90	2	1
12	0.80	2	2
13	1.10	2	3
14	0.10	2	4
15	-0.10	2	5
16	4.40	2	6
17	5.50	2	7
18	1.60	2	8
19	4.60	2	9
20	3.40	2	10

Another example

	extra	group	ID
1	0.70	1	1
2	-1.60	1	2
3	-0.20	1	3
4	-1.20	1	4
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8	0.80	1	8
9	0.00	1	9
10	2.00	1	10
11	1.90	2	1
12	0.80	2	2
13	1.10	2	3
14	0.10	2	4
15	-0.10	2	5
16	4.40	2	6
17	5.50	2	7
18	1.60	2	8
19	4.60	2	9
20	3.40	2	10

- `extra` seems to be numerical or continuous
- `group` numerical, categorical or binary are all possible options (without context)

Descriptive Statistics

- It is important to investigate what each variable contains - performing **Exploratory Data Analysis** or **EDA**
- There are different ways of describing or exploring each variable and the approach will depend on their type: categorical vs numerical
- In each case it is possible to display information about the variable as a numerical summary or a graph
- It is also possible to look at how variables relate to each other, but we will cover this later in the module

The phrase *exploratory data analysis* was coined by John W. Tukey.

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Categorical Data - Descriptive Statistics

There are different ways to numerically summarise a **categorical** variable:

- Frequencies or counts
- Relative Frequencies
- Relative Cumulative Frequencies

Categorical Data - Descriptive Statistics

Frequency table for `Vegetation`

	Frequency
Arable	3
Grassland	9
Meadow	3
Orchard	1
Scrub	4

Categorical Data - Descriptive Statistics

Frequency table for `Vegetation`

	Frequency	Cumulative Frequency
Arable	3	3
Grassland	9	?
Meadow	3	?
Orchard	1	?
Scrub	4	?

What are the values to replace ?

Categorical Data - Descriptive Statistics

Frequency table for `Vegetation`

	Frequency	Cumulative Frequency
Arable	3	3
Grassland	9	12
Meadow	3	15
Orchard	1	16
Scrub	4	20

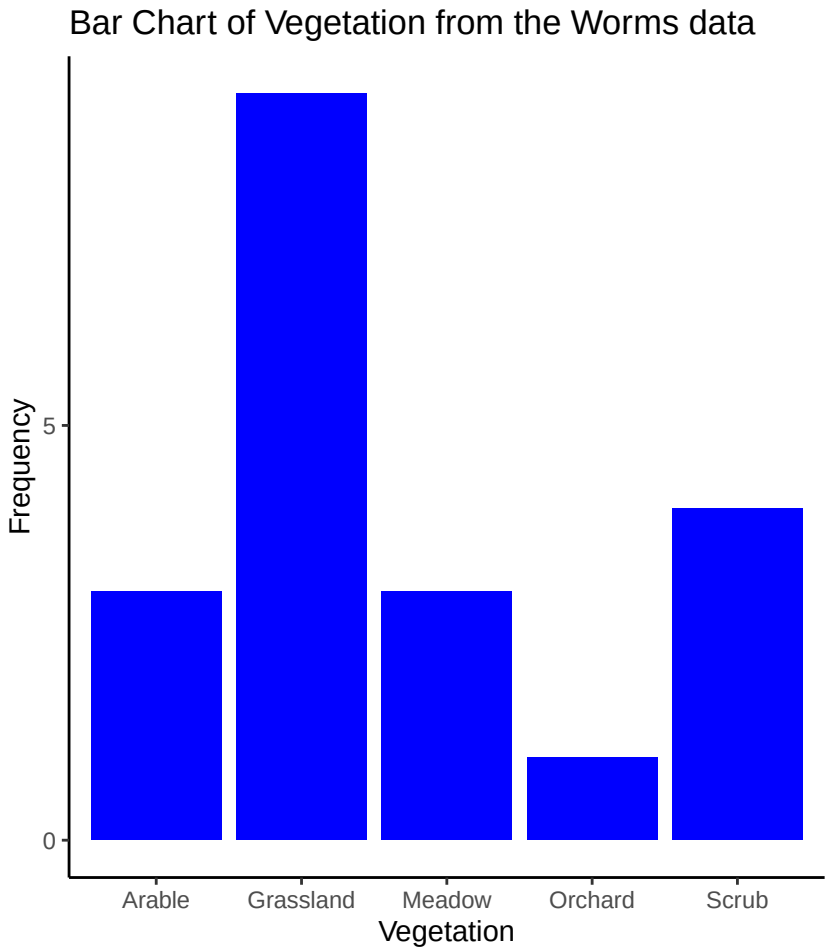
Categorical Data - Visualising

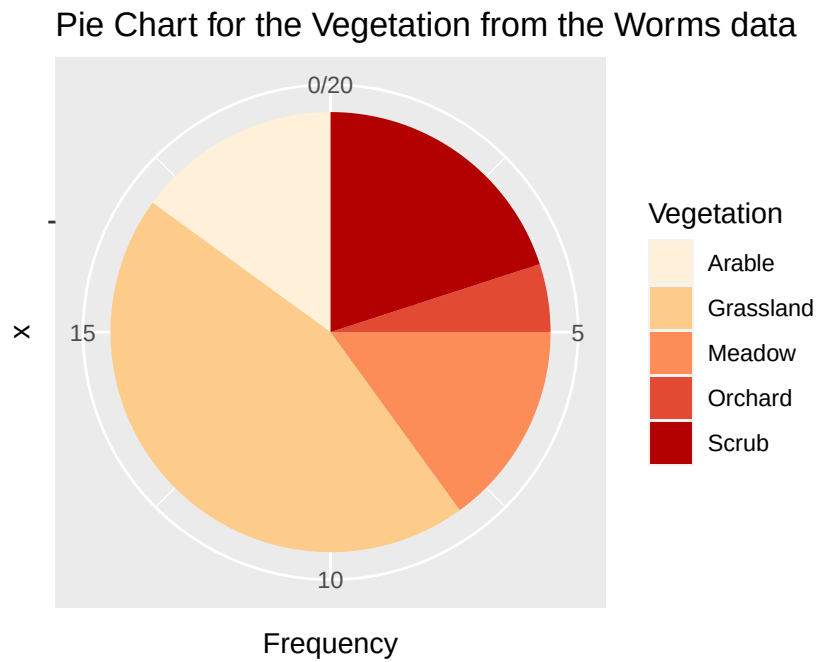
Categorical Variables can be summarised visually using; **Bar Charts** or **Pie charts**

- **Categorical** Variables can be summarised visually using; **Bar Charts** or **Pie charts**

- **Bar Charts** can be used to represent the frequencies of each of the different categories, The y-axis has the frequency and the x-axis the categories.
- **Pie Charts** can be used to represent the frequencies of each of the different categories as different slices of the pie.

Categorical Data- Visualising





Numerical Data - Descriptive Statistics

When describing the contents of a numerical variable we can look at different aspects of its distribution, such as

- Measures of location - such as the mean
- Measures of spread and variability
- Extreme values

Numerical Data - Descriptive Statistics - Measures of Location

- **Mode** the most frequent observation in the data set, there can be more than one
- **Mean** (average) this is calculated as the sum of the observations in the variable divided by the number of the observations.

$$\bar{x} = \frac{1}{n}\Sigma_{i=1}^{n} x_i$$

assuming we have an attribute X which has the following observations $x_1, x_2, \ldots, x_n$

- **Median** when the set of observations is ordered from smallest to largest, the **median** is the value in the middle, or if there are an even number of observations then the mean of the two middle observations is used.

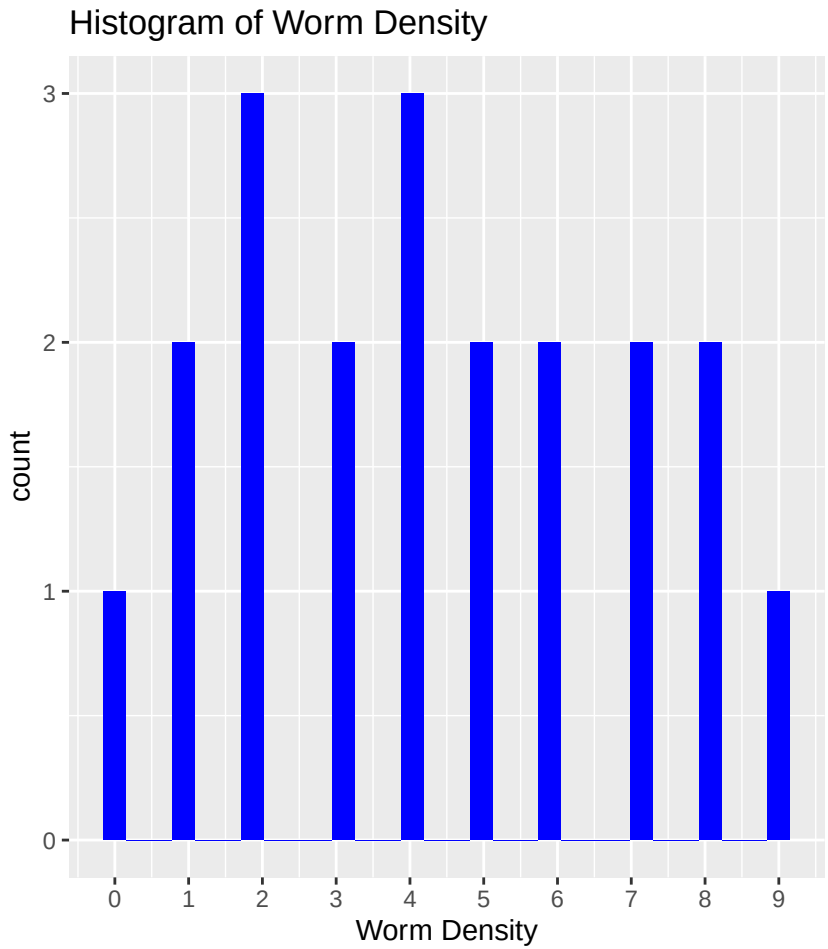
Note: the **mean** is affected by extreme values, the **median** is not.

Numerical Data - Descriptive Statistics - Measures of Variation

There are different ways of quantifying the spread:

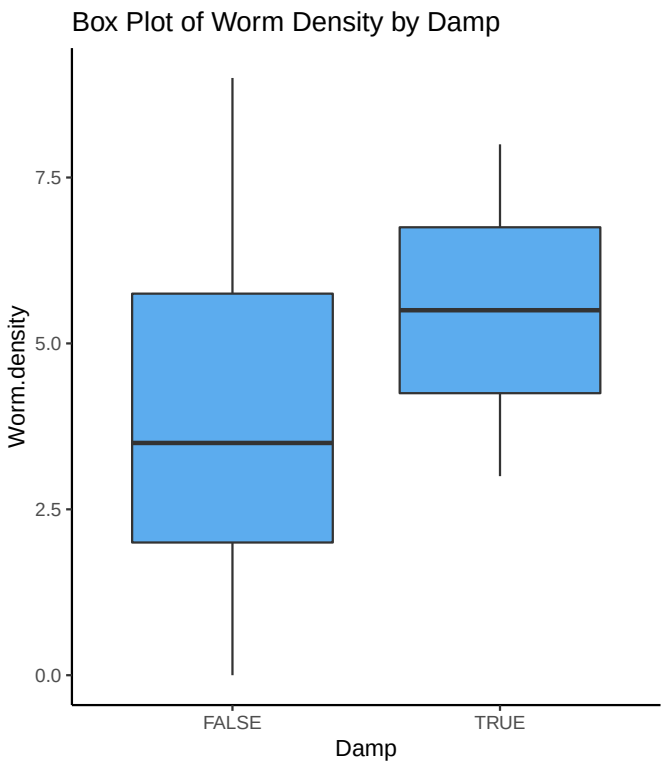
- The **range**: defined as *maximum value in the variable - minimum value*
- The **Inter Quartile Range - IQR** defined as the *75th Percentile - 25th Percentile*, this will measure the spread of the central 50% of the data
- The **Standard Deviation** - measures the spread of the observations from the mean

Numerical Data - Visualising - Histogram



- A **Histogram** shows the frequency of values in the data.
- For example: the graph on the left, we can read that:
  - In the data there is only one garden with a worm density of 9
  - The most frequent values are: 4 and 2. They each appear 3 times in the data.

Numerical Data - Visualising - Box Plot



- A **Box Plot** or **Box and Whisker** Plot displays the data using the median, the 25th and 75th percentile and the minimum and maximum.
  - A **Box Plot** can also display extreme observations or outliers
  - The middle of the box is the *median*
  - The edges of the box are the 25th and 75th percentiles
- In the example to the left:
  - the median `worm density` when `Damp` is `TRUE` is higher then when it is `FALSE`
  - the spread of the data is larger when `Damp` is `FALSE`

Summary

In this lecture:

- Why is statistics important
- Outlined the different types of variables
- Introduced EDA and methods for descriptive statistics for different types of variables

In the Labs - Using RStudio to read in data, explore the data with numerical summaries and graphs.

QDA - What's next?

- This afternoon in the labs you will have a chance to do this independently
- We will be around to answer questions
- Look out for the material for next week where we will focus on confidence intervals